

Viscoelastic Memory Kernels and Cosmological Transport Transitions in Geotopological Hydrodynamics

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ABSTRACT

We formalize the non-linear viscoelastic memory transport and cosmological expansion mechanism in Geotopological Hydrodynamics (GTH v12.0), showing how cosmic acceleration, sound horizon shifts, and redshift-space attenuation emerge without dark energy scalar fields or cosmological constants Λ . We analyze the Israel-Stewart causal relaxation transport equation $\tau_0 \Delta^\mu_\alpha \Delta^\nu_\beta u^\lambda \nabla_\lambda \Pi^{\alpha\beta} + \Pi^{\mu\nu} = -2 \eta_{\text{phase}}^{\text{eff}}(x) \sigma_n^{\mu\nu}$, where $\eta_{\text{phase}}^{\text{eff}} = \eta_n \chi_n f_n$, and prove its exact equivalence to the retarded memory kernel representation $\Pi_n^{\mu\nu}(t) = -(2\eta_n/\tau_0) \int_{-\infty}^t e^{-(t-s)/\tau_0} \chi_n(s) f_n(s) \sigma_n^{\mu\nu}(s) ds$. We evaluate the GTH Friedmann-like cosmological expansion law $H^2 = (8\pi G_{\text{model}}/3) (\rho_{\text{GK}} + \rho_{\text{phonons}}) - k c_{\text{SI}}^2/a^2 + \Lambda_{\text{curv}} c_{\text{SI}}^2/3$, deriving line-of-sight attenuation $H_{\text{obs}}(r) = H_{\text{bg}}(r) + \alpha_{\text{att}}(r)$ with composite transport rate $Y_{\text{tr}}(r) \equiv v_{\text{tr}}(r)/l_c(r)^2$ and projection weight $\delta_\Omega = 1/64$. We prove the Pre-Recombination Stability Theorem: for $u(z) \equiv Y_{\text{tr}}(z)/H(z) \ll 1$, homogeneous density modes redshift as $\rho_{\text{GK}} \propto a^{-3}$ and $\rho_{\text{phonons}} \propto a^{-4}$, preserving standard CMB acoustic peak ratios and the BAO acoustic scale $r_s(z_*) \approx 147$ Mpc. Finally, we derive the differential redshift-distance inversion $dr = c_s dz / [(1+z) H_{\text{obs}}(z)]$ and the Müller-Israel-Stewart causal viscosity limit $\eta_n/\eta_{\text{causal}} \leq 1$ with $\eta_{\text{causal}} = \tau_0 \rho_0 c_{\text{substrate}}^2 = 1.0100 \times 10^{-12}$ Pa s.

1. INTRODUCTION & THE COSMOLOGICAL MEMORY PARADIGM

In Standard Lambda-CDM cosmology, late-time cosmic acceleration is attributed to an uncalculated vacuum energy density Λ_{CDM} . In Geotopological Hydrodynamics (GTH v12.0), cosmological transport and expansion emerge as the non-local, retarded viscoelastic relaxation of the 5D superfluid substrate.

All transport is governed by the single constitutive state vector $\Theta = (M_{\text{UV}}, m_{\text{IR}}, \rho_0, K, G_{\text{shear}}, \tau_0, \eta_n)$, with $\tau_0 = 1.2500 \times 10^{-2}$ s, $\rho_0 = 1.0100 \times 10^{-26}$ kg m⁻³, and $G_{\text{shear}} = 8.0797 \times 10^{-11}$ Pa.

2. ISRAEL-STEWART RELAXATION & RETARDED MEMORY KERNEL

The causal relaxation equation for the non-ideal shear stress tensor $\Pi^{\mu\nu}$ is:

$$\tau_0 \Delta^\mu_\alpha \Delta^\nu_\beta u^\lambda \nabla_\lambda \Pi^{\alpha\beta} + \Pi^{\mu\nu} = -2 \eta_n \chi_n f_n \sigma_n^{\mu\nu}$$

THEOREM 1 (Retarded Memory Kernel Equivalence).

For any linearized forcing history with vanishing past stress $\Pi^{\mu\nu}(-\infty) = 0$, the Israel-Stewart relaxation law is exactly equivalent to the retarded integral memory kernel:

$$\Pi_n^{\mu\nu}(t) = -\frac{2\eta_n}{\tau_0} \int_{-\infty}^t e^{-(t-s)/\tau_0} \chi_n(s) f_n(s) \sigma_n^{\mu\nu}(s) ds$$

Proof: Integrating the first-order differential equation $\tau_0 d\Pi/dt + \Pi = -2 \eta_{\text{phase}}^{\text{eff}} \sigma_n$ using integrating factor e^{t/τ_0} yields $\Pi(t) = e^{-t/\tau_0} \int_{-\infty}^t e^{s/\tau_0} (-2 \eta_{\text{phase}}^{\text{eff}}(s) \sigma_n(s) / \tau_0) ds$. Grouping $e^{-(t-s)/\tau_0}$ inside the integrand yields the exact retarded kernel. Q.E.D.

3. COSMOLOGICAL EXPANSION & LINE-OF-SIGHT ATTENUATION

The FLRW reduction of the GTH field equations yields the cosmological Friedmann analogue:

H^2(a) = \frac{8\pi G_{model}}{3} (\rho_{GK} + \rho_{phonons}) - \frac{kc_{SI}^2}{a^2} + \frac{\Lambda_{curv} c_{SI}^2}{3}

Observed expansion along line-of-sight propagation paths incorporates dissipative attenuation:

H_{obs}(z) = H_{bg}(z) + \delta_{\Omega} Y_{tr}(z), \quad \delta_{\Omega} \equiv \frac{1}{64}, \quad Y_{tr}(z) \equiv \frac{v_{tr}(z)}{l_c(z)^2}

4. PRE-RECOMBINATION EIGENMODE STABILITY & BAO SOUND HORIZON

Coupled continuity equations for Geo-Knot density \rho_{GK} and phonon density \rho_{phonons} yield a 2x2 transport matrix:

\lambda_{\pm} = \frac{-7 - 2u \pm \sqrt{1 + 4u^2}}{2}, \quad u(z) \equiv \frac{Y_{tr}(z)}{H(z)} \ll 1 \Rightarrow \rho_{GK} \propto a^{-3}

THEOREM 2 (Pre-Recombination Eigenmode Conservation).
In the weak-mixing limit u(z) \equiv Y_{tr}(z)/H(z) << 1 before recombination, the matrix eigenvalues reduce to \lambda_{-} = -3 and \lambda_{+} = -4, preserving standard matter (\rho \propto a^{-3}) and radiation (\rho \propto a^{-4}) scaling.

The BAO acoustic sound horizon integral uses the baryon-photon sound speed c_{s,\gamma b}(z):

r_s(z^*) = \int_{z^*}^{\infty} \frac{c_{s,\gamma b}(z) dz}{H_{bg}(z) + \delta_{\Omega} Y_{tr}(z)} \approx 147 \text{ Mpc}

5. MÜLLER-ISRAEL-STEWART CAUSAL VISCOSITY CEILING

Causal characteristic signal propagation requires the viscosity ceiling:

c_{visc}^2 = \frac{\eta_{dyn}}{\tau_0(\rho + p/c_{substrate}^2)} < c_{substrate}^2 \Rightarrow \eta_{causal} = \tau_0 \rho_0 c_{substrate}^2 = 1.0100 \times 10^{-12} \text{ Pa s}

6. CONCLUSION & MONOGRAPH SERIES ROADMAP

Viscoelastic memory kernels generate cosmological transport and redshift attenuation without dark energy, maintaining pre-recombination stability u(z) << 1 and BAO sound scale r_s \approx 147 Mpc.

Paper #	Monograph Title / Physical Focus	Key Formulation
Paper 1	Density Ceiling & Core Regularization	V_{top}(\rho) \rightarrow +\infty, R_c \geq (3M_c / 4\pi \rho_{max})^{1/3}
Paper 2	Scale Isolation & Chiral Orthogonality	M_{\{G,eff\}}^2 = M_G^2 / (1 + Z(\text{grad } \phi))
Paper 3	Beltrami Vortex Wakes & Disk Dynamics	\beta(r) = \chi_n G_{model} \Sigma_b(r) e^{-(\text{Re } GTH)}
Paper 4	Acoustic Horizons & Echo Comb	\tau_{echo}(\omega) = 2 (d \Phi_{sv} / d \omega)
Paper 5	Zero-Mode Gauss-Codazzi Projection	G^{(4)}_{\{\mu \nu\}} + \Lambda_{curv} h_{\{\mu \nu\}} = \kappa_{SI} T^{\{eff\}}_{\{\mu \nu\}} + C_{\{\mu \nu\}}
Paper 6	Viscoelastic Memory & Cosmological Transport	H_{obs}(z) = H_{bg}(z) + \delta_{\Omega} Y_{tr}(z), r_s \approx 147 \text{ Mpc}